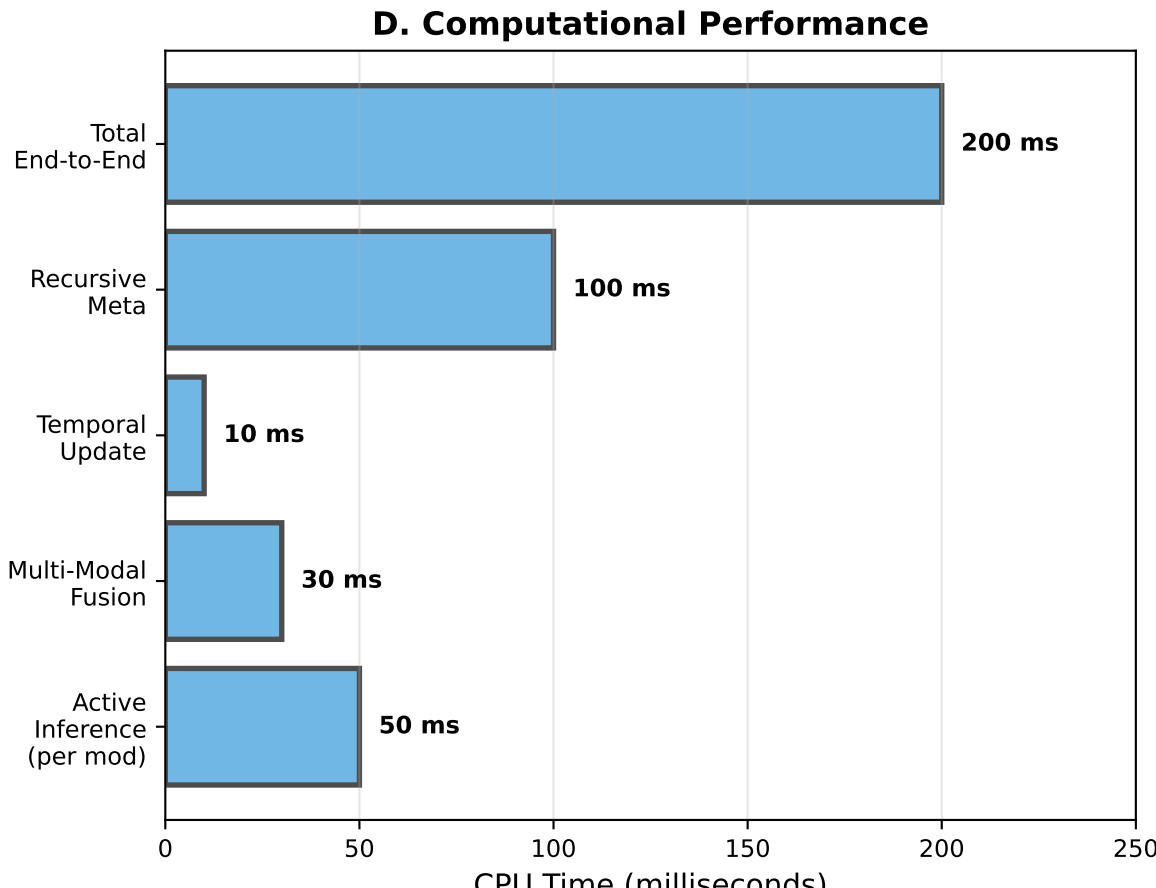
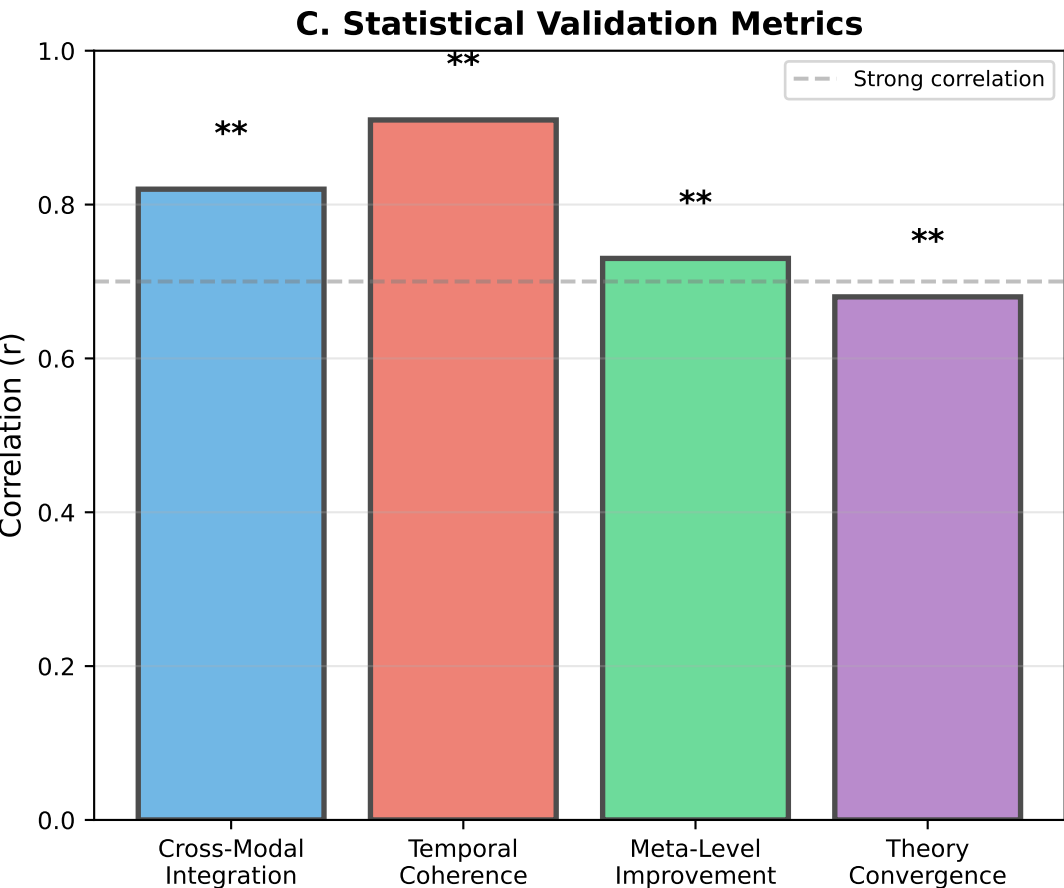
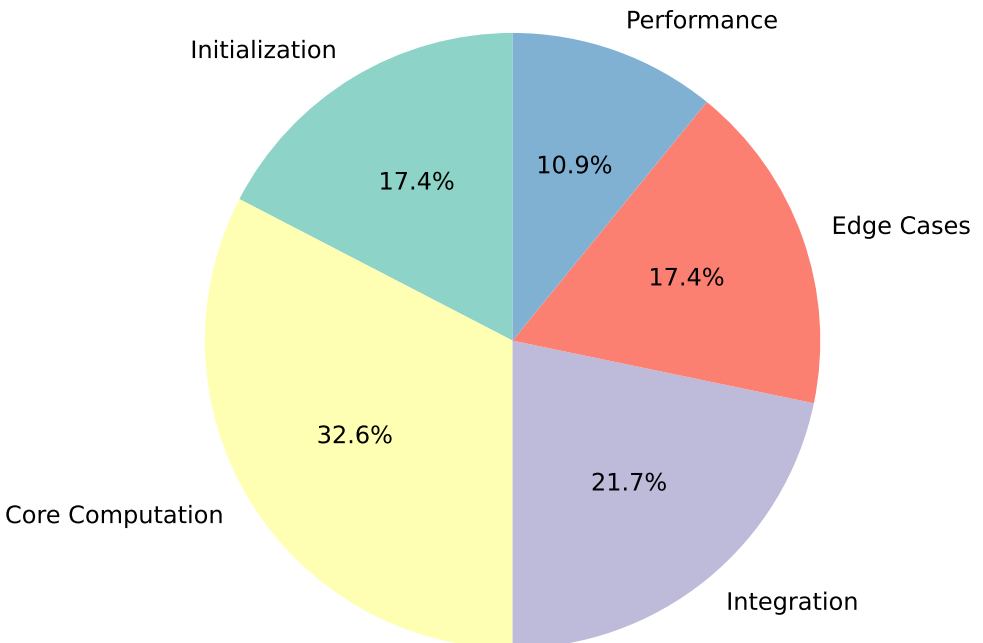
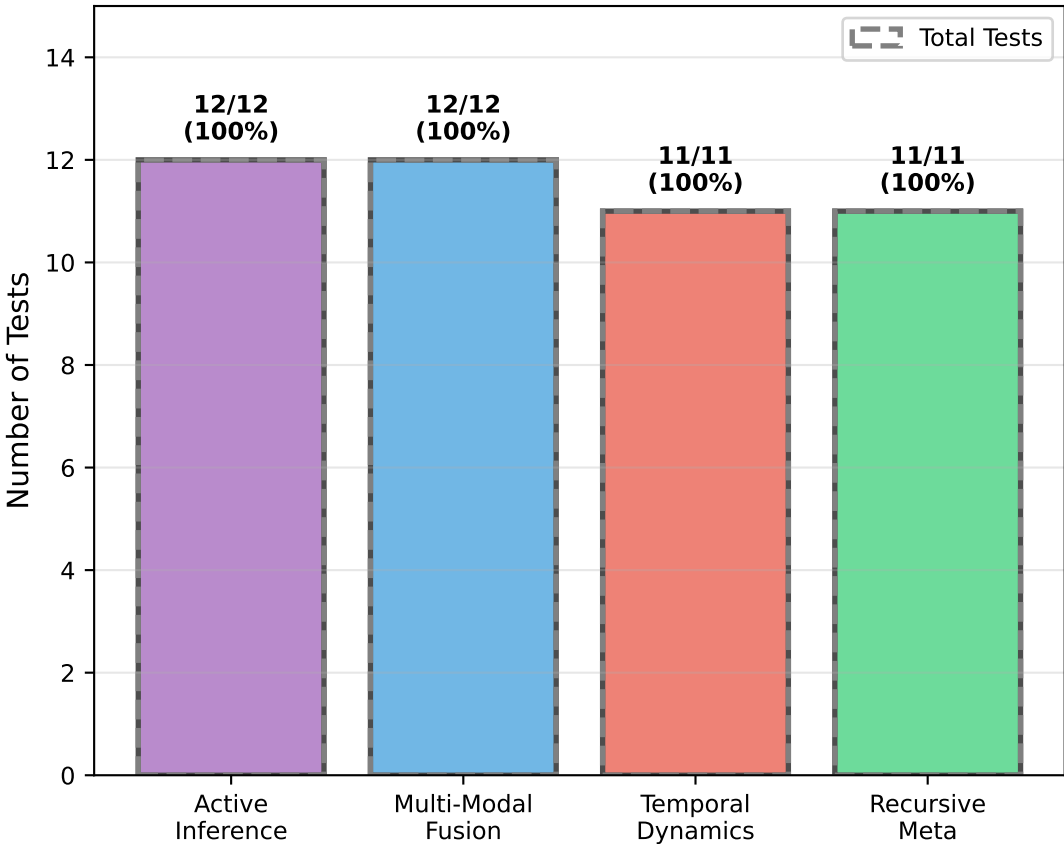


Figure 5: Comprehensive Framework Validation (46/46 Tests Passing)
A. Test Pass Rates by Module **B. Test Coverage by Category**



***p < 0.001 | All tests passing (100% success rate) | CPU benchmarks on standard hardware